

risk-adjusted opportunities are; it only provides a method of systematically evaluating them. I cover a promising area of alpha in section 4.

3. Factor Benchmarks

Consider the CAPM applied to asset (or strategy or fund) i :

$$E(r_i) - r_f = \beta_i(E(r_m) - r_f), \quad (10.6)$$

where $E(r_i)$ is the expected return of asset i , r_f is the risk-free rate (U.S. T-bills), β_i is the beta of asset i , and $E(r_m)$ is the expected return of the market portfolio. Let's assume the beta is 1.3, $\beta_i = 1.3$, so we can write equation (10.6) as

$$E(r_i) = r_f + 1.3E(r_m) - 1.3r_f,$$

and, rearranging, we have

$$\underbrace{E(r_i)}_{\$1} = \underbrace{-0.3r_f + 1.3E(r_m)}_{\$1}. \quad (10.7)$$

Applying the CAPM assumes that we can produce the same return as \$1 in asset i by holding a short position in T-bills of $-\$0.30$ and a levered position in the market of $\$1.30$. Note that we have \$1 on both the left-hand side and right-hand side of equation (10.7). Thus, the CAPM implies a *replicating portfolio*: a combination of risk-free assets and the market has the same expected return as the asset under the CAPM. Thus, the beta of the asset is actually a *mimicking portfolio weight*.⁶

Factor benchmarks are a combination of investment portfolios, or factors, on the right-hand side that give the same return as the asset on the left-hand side. The factor benchmark describes the systematic components of asset i 's return.

The alpha of asset i is any expected return generated in excess of the short \$0.30 in T-bills and the long \$1.30 position in the market portfolio:

$$E(r_i) = \alpha_i + \underbrace{[-0.3r_f + 1.3E(r_m)]}_{E(r^{bmk})}, \quad (10.8)$$

where the benchmark consists of the risk-adjusted amount held in equities and the risk-free rate, $r^{bmk} = -0.3r_f + 1.3r_m$. Since the benchmark in this case comes from the CAPM, alpha is the average return in excess of the return predicted by the CAPM.

⁶Equation (10.7), which can be estimated in a regression (see below), takes the perspective of matching \$1 invested in the asset on the left-hand side with the factor benchmark on the right-hand side. An alternative approach is to scale both the left- and right-hand sides so that they have the same volatility. This would match risk exposures rather than dollars invested.